MediaMix: Multimedia Retrieval with Dual Backend Support and Result Exploration in MR

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Abstract. Extended Reality (XR), particularly Mixed Reality (MR), offers exciting opportunities for immersive user interactions that exceed the limitations of traditional two-dimensional interfaces. Following our first participation in the Video Browser Showdown (VBS) 2025, we present an extended and improved version of MediaMix, our mixed reality multimedia retrieval system running on the Apple Vision Pro. This new iteration introduces substantial enhancements to both the backend and the frontend. On the backend, MediaMix now seamlessly integrates with FERElight and the vitrivr-engine, thereby extending its flexibility in retrieval workflows and ensuring compatibility with established infrastructures within the vitrivr stack. On the frontend, we expanded the presentation of results to enhance the browsing experience. Beyond the existing spherical visualization, results can now be displayed in floating windows or in wall-mounted views, giving users greater control over their workspace arrangement. Furthermore, we extended the sphere-based representation, allowing users to freely configure the number of displayed results. Moreover, multiple spheres can be merged into combined result sets, enabling a more comprehensive and comparative exploration of search outcomes. With these enhancements, we aim to increase usability, adaptability, and efficiency of MediaMix in interactive video retrieval tasks.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Mixed Reality

1 Introduction

Mixed Reality (MR) [10] appears as a powerful paradigm within the broader field of Extended Reality (XR), combining the immersive nature of Virtual Reality (VR) with the contextual embedding of Augmented Reality (AR). By seamlessly blending digital and physical objects in a single environment, MR enables novel forms of interaction that go beyond the limitations of conventional two-dimensional interfaces. These capabilities open up new possibilities across diverse

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domains, including education, healthcare, entertainment, and remote collaboration. In the area of multimedia retrieval, and in particular interactive video retrieval, MR holds the potential to transform user interaction by supporting natural query input, immersive exploration, and spatial result visualization. The Video Browser Showdown (VBS) [8] provides a competitive framework for exploring these innovations. As an international annual challenge, VBS tasks participants with retrieving specific video segments from large-scale datasets under strict time constraints. The competition includes multiple query types: known-item search (KIS), where the usesers must found a unique target segment based on a visual or textual description; ad-hoc video search (AVS), in which participants retrieve as many relevant segments as possible for a natural language query; and visual question answering (VQA), where systems must extract factual information from video data to answer questions. These tasks are performed on heterogeneous datasets such as the V3C collection [9], the LHE75 laparoscopic surgery dataset [4], and the Marine Video Kit [11], ensuring that systems are evaluated under diverse retrieval scenarios. The VBS utilizes DRES [7] for task presentation, as well as submission collection and evaluation.

Recent advances in XR hardware provide the technological foundation for bringing video retrieval into immersive environments. Devices such as the Apple Vision Pro[1], XReal Glasses[2], and Meta Quest 3[3] demonstrate how lightweight, high-resolution, and spatially aware headsets can support intuitive gesture-based and gaze-based interaction. These advances open opportunities for multimedia retrieval applications that were previously constrained to desktop or completely virtual reality settings. In this paper, we present the new version of MediaMix [1], a mixed reality multimedia retrieval system designed for the Apple Vision Pro and extended beyond its 2025 prototype. Building on our earlier work, we introduce enhancements to both the backend and frontend. On the backend, MediaMix now integrates with FERELight[4] and the vitrivr engine [2], thereby extending retrieval flexibility and compatibility with the vitrivr stack, where a new API has been employed. On the frontend, we expand the result visualization modes: in addition to the immersive spherical representation, users can now display results in floating or wall-mounted windows, configure the density of results on spheres, and merge multiple spheres into combined result sets. Together, these improvements provide users with greater flexibility in formulating, arranging, and exploring queries and results within a mixed reality environment. Therefore, we aim to gather insights into the browsing experience by using the various result window options.

With these contributions, MediaMix aims to demonstrate how MR can enhance usability, adaptability, and efficiency in interactive video retrieval. The system will participate in VBS 2026, where its new features will be used to complete time-critical retrieval tasks in large and diverse datasets.

1 https://www.apple.com/apple-vision-pro.
2 https://next.xreal.com/air2ultra.
3 https://www.meta.com/ch/en/quest/quest-3.
4 https://github.com/FEREorg/ferelight.

The remainder of the paper is structured as follows: Sect. 2 introduces the MediaMix system. Section 3 summarizes the extensions to MediaMix for VBS 2026 and Sect. 4 concludes.

2 MediaMix System Overview

MediaMix builds on its first prototype version, which was used at VBS 2025, introducing several significant enhancements that improve both functionality and usability. It features a modular architecture that combines a retrieval backend with a mixed reality (MR) frontend. Currently, vitrivr-engine and FERElight can both be selected as the retrieval engine. The backend is in charge of scalable and efficient query execution, while the frontend prioritizes an intuitive user experience through natural input methods and spatial exploration of results in MR. MediaMix operates on the Apple Vision Pro, leveraging its high-resolution display and capabilities for gaze- and gesture-based interaction, as well as keyboard connectivity. Furthermore, the system offers a connection to DRES. This integration allows users to submit query results directly from within the MR user interface. Figure 1 provides an overview of the overall system architecture.

3 MediaMix System Extensions

In this section, we summarize the extensions to MediaMix for the 2026 VBS participation.

3.1 Backend

We extended the backend of MediaMix substantially compared to its 2025 iteration, thereby improving both functionality and versatility. While the previous version was based exclusively on the FERElight backend, the current system introduces support for multiple retrieval engines by also incorporating vitrivr-engine. This dual-backend architecture broadens the methodological scope of MediaMix and enables more flexible retrieval workflows.

- The vitrivr-engine represents the most recent retrieval engine of the vitrivr stack and provides advanced functionality for multimedia indexing, feature extraction, and retrieval. It supports a wide range of multimodal features, including Optical Character Recognition (OCR) for text extraction, Automatic Speech Recognition (ASR) for transcribed speech, and modern embedding techniques derived from models such as CLIP [6] and DINOv2 [5]. Its modular query pipelines enable the construction of tailored search processes. We plan to use these four modalities at VBS 2026. For database support, we connect vitrivr-eninge with PostgreSQL[5], including the pgvector extension[6], which enables efficient similarity search.

[5] https://www.postgresql.org.
[6] https://github.com/pgvector/pgvector.

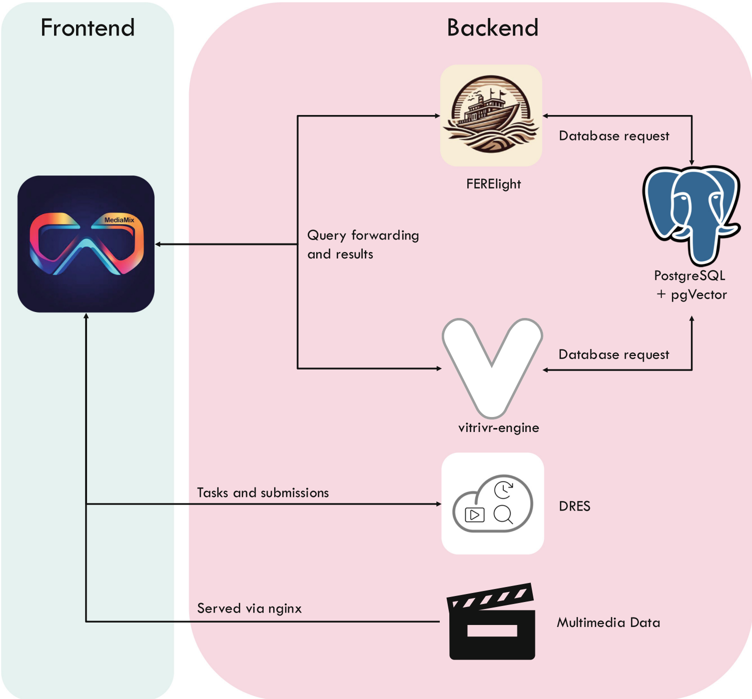

Fig. 1. Architecture of MediaMix, showing the connection between the backends (vitrivr-engine and FERElight), the frontend in mixed reality, and the DRES evaluation server.

– In contrast, FERElight is a lightweight, Python-based backend that was originally developed for VBS 2025. It focuses on speed and efficiency. FERElight provides a practical solution in cases where reduced complexity and rapid responses are prioritized over advanced multimodal processing. It offers support for simple CLIP, OCR, and ASR queries.

By supporting both FERElight and vitrivr-engine, MediaMix achieves a high degree of adaptability. Users can choose between lightweight and advanced retrieval strategies depending on the task at hand, ensuring that MediaMix is fitted to a comprehensive range of use cases.

3.2 Frontend

The frontend of MediaMix builds the core of the mixed reality experience. We extended it compared to the system with which we participated in the last iteration. While the earlier version primarily focused on a prototype retrieval system and the use of spheres for result visualization, the current system extends this approach. It introduces new modes of interaction and result visualization to enhance usability and adaptability. Spheres are still playing a central role

(a) Results displayed on spheres with configurable density.

(b) Results displayed in two-dimensional panel.

Fig. 2. Result visualization in MediaMix using different display modes.

in presenting retrieval results, but we expanded their functionality. Instead of being limited to a fixed number of results previewed, users can now configure the resolution of each sphere. Therefore, we can create a range of visualizations, from compact overviews to more detailed explorations. This flexibility enables the visualization to adapt to various query requirements, providing more detailed control over the level of detail within the retrieval workspace. Spheres with different resolutions are shown in Fig. 2a.

Furthermore, we integrated a t-SNE–based arrangement of results to improve spatial organization within the spheres. In the previous version, items on a sphere were displayed strictly according to their retrieval score, which limited the ability to perceive relationships between results. With the new approach, items are instead arranged according to their positions in a t-SNE embedding [3], grouping similar results closer together and forming visually coherent clusters. This clustering allows users to identify patterns, such as groups of semantically related results. We include this option to provide a more intuitive exploration of the retrieval space. Another significant extension is the introduction of merged spheres. Multiple spheres belonging to distinct queries can be combined into a single integrated result set. This feature is particularly beneficial when queries span different modalities, as it enables users to examine results in a unified space without losing contextual information.

In addition to spheres, results can now be visualized on two-dimensional panels (see Fig. 2b). These panels can either be placed freely in the mixed reality environment or anchored to a virtual wall, providing alternative arrangements for organizing and comparing search results. The panel-based view offers a more traditional presentation style, complementing the immersive character of the spheres and enabling parallel exploration of multiple queries.

Interaction with these visualizations is still facilitated through the gaze- and gesture-based interaction capabilities of the Apple Vision Pro. Users can rotate, reposition, or resize objects naturally, and switch seamlessly between different visualization modes. Selecting a result opens a playback window that supports detailed inspection, including frame-level navigation and the exploration of related segments. Together, these interaction mechanisms provide an immersive and fluid workflow for multimedia retrieval, enabling users to progress seamlessly from query formulation to result evaluation entirely within the mixed reality environment.

4 Conclusion

With MediaMix, we extend mixed reality multimedia retrieval towards a more flexible and immersive system. By introducing support for both FERElight and vitrivr-engine, we enable adaptable retrieval workflows ranging from lightweight to advanced multimodal search. On the frontend, we expanded the visualization beyond spherical representations, allowing for configurable densities, merged spheres, clustered ordering, and alternative 2D panel-based views. These enhancements provide users with more control over their retrieval workspace. Our goal is to improve usability and efficiency. We will evaluate its performance in a competitive environment at VBS 2026.

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References

1. Arnold, R., Kempf, R., Waltenspül, R., Schuldt, H.: MediaMix: multimedia retrieval in mixed reality. In: Ide, I., et al. (eds.) MultiMedia Modeling. MMM 2025. LNCS, vol. 15524, pp. 310–317. Springer, Singapore (2025). https://doi.org/10.1007/978-981-96-2074-6_37
2. Gasser, R., Arnold, R., Faber, F., Schuldt, H., Waltenspül, R., Rossetto, L.: A new retrieval engine for Vitrivr. In: Rudinac, S., Hanjalic, A., Liem, C., Worring, M., Jónsson, B.Þ., Liu, B., Yamakata, Y. (eds.) MultiMedia Modeling, LNCS, pp. 324–331. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-53302-0_28
3. van der Maaten, L., Hinton, G.: Visualizing data using t-sne. J. Mach. Learn. Res. **9**(86), 2579–2605 (2008). http://jmlr.org/papers/v9/vandermaaten08a.html
4. Nasirihaghighi, S., et al.: Gynsurg: a comprehensive gynecology laparoscopic surgery dataset (2025). https://arxiv.org/abs/2506.11356
5. Oquab, M., et al.: DINOv2: learning robust visual features without supervision. CoRR **abs/2304.07193** (2023)
6. Radford, A., et al.: Learning transferable visual models from natural language supervision (2021). https://arxiv.org/abs/2103.00020
7. Rossetto, L., et al.: On the user-centric comparative remote evaluation of interactive video search systems . https://doi.org/10.1109/MMUL.2021.3066779, https://dres.dev
8. Rossetto, L., Schoeffmann, K., Gurrin, C., Lokoč, J., Bailer, W.: Results of the 2025 video browser showdown (2025). https://arxiv.org/abs/2509.12000

9. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C – a research video collection. In: MultiMedia Modeling, LNCS, vol. 11295, pp. 349–360. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-05710-7_29
10. Speicher, M., Hall, B.D., Nebeling, M.: What is mixed reality? In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, pp. 1–15. CHI '19, Association for Computing Machinery, New York, NY, USA (2019). https://doi.org/10.1145/3290605.3300767
11. Truong, Q., et al.: Marine video kit: a new marine video dataset for content-based analysis and retrieval. In: Dang-Nguyen, D., et al. (eds.) MultiMedia Modeling - 29th International Conference, MMM 2023, Bergen, Norway, 9–12 January 2023, Proceedings, Part I. LNCS, vol. 13833, pp. 539–550. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-27077-2_42